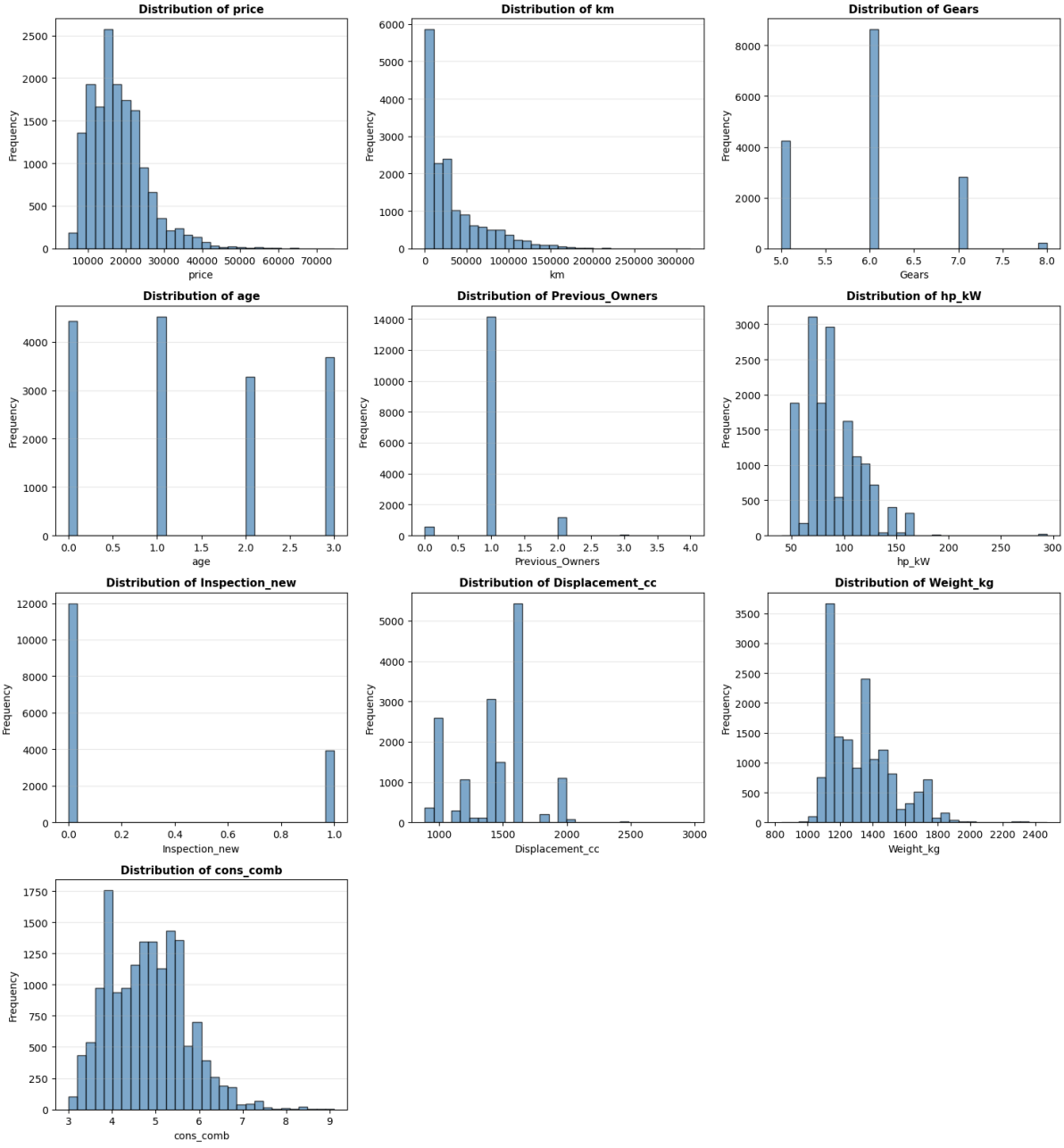
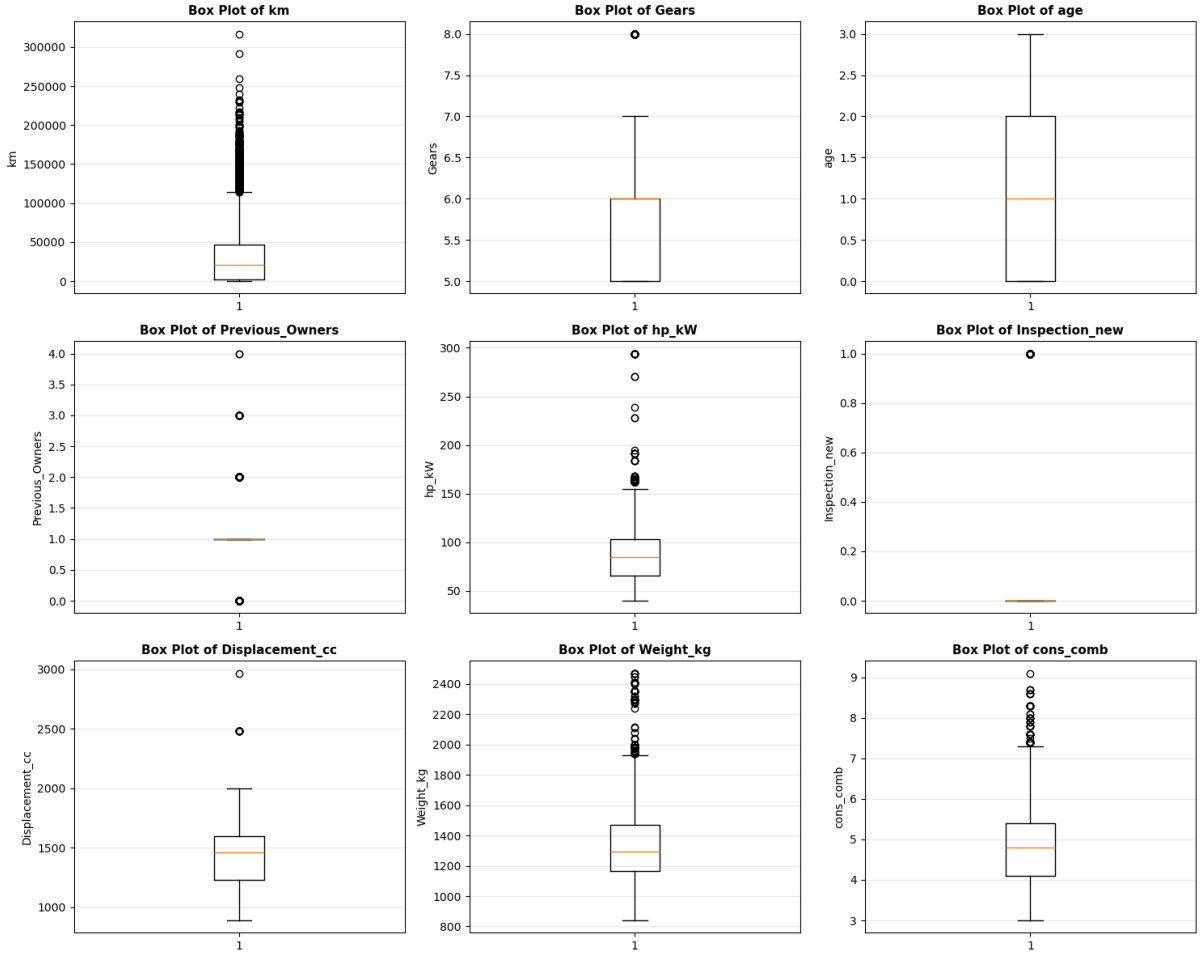
Preliminary Analysis and Frequency Distributions

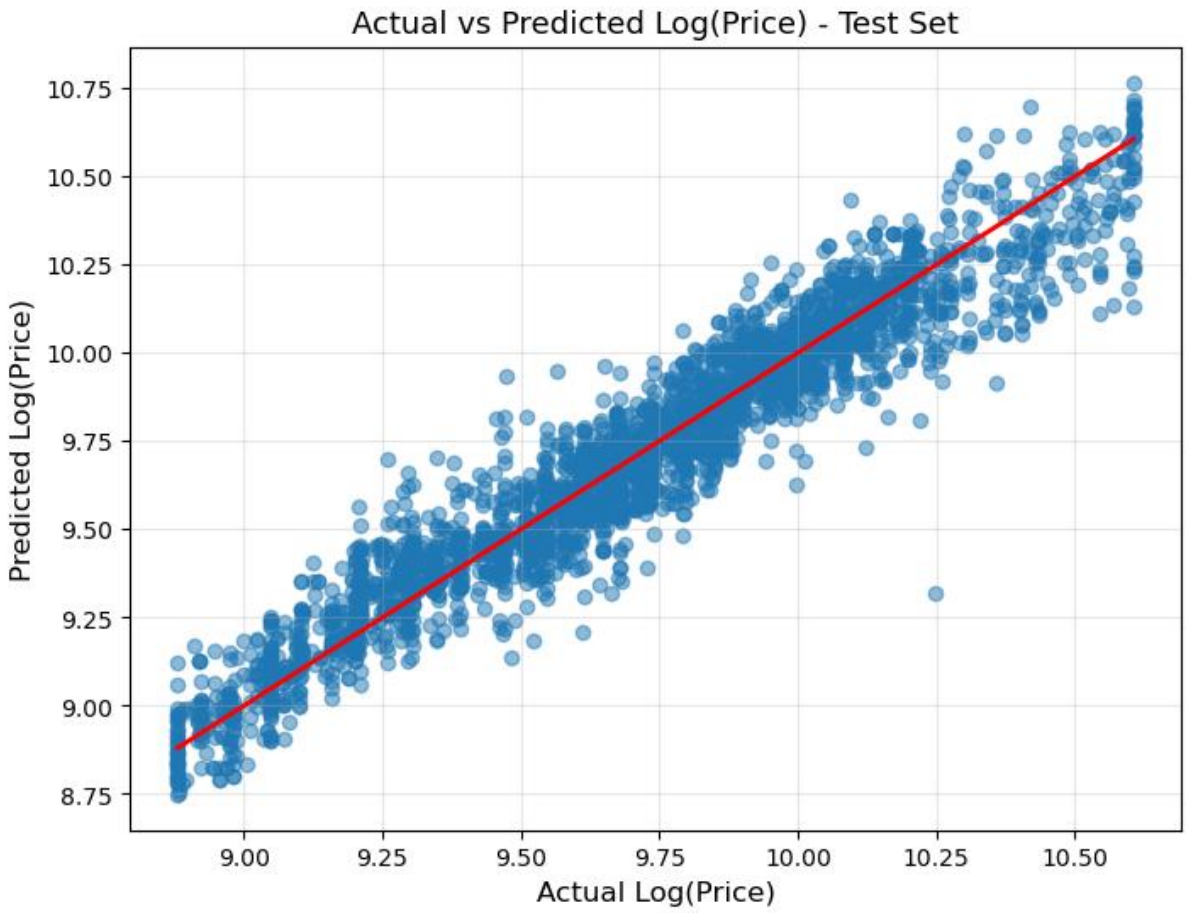


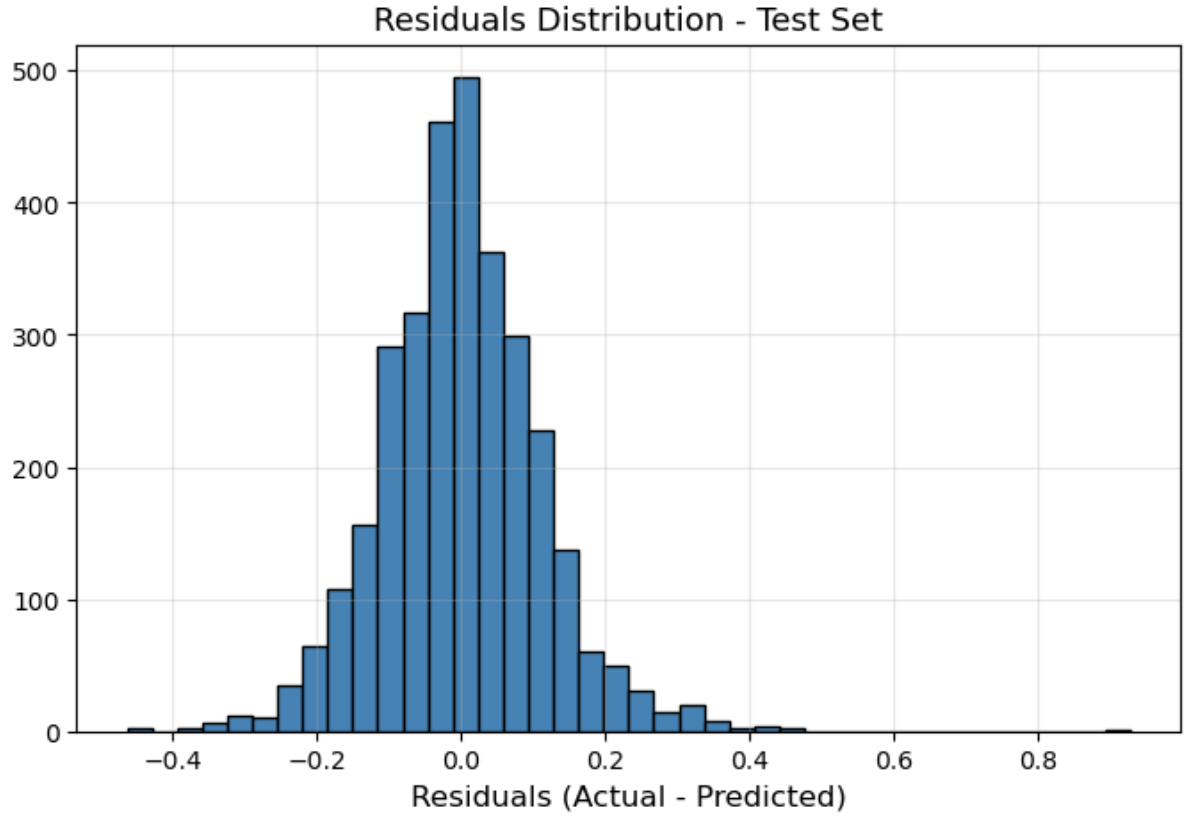
- .

3. Exploratory Data Analysis

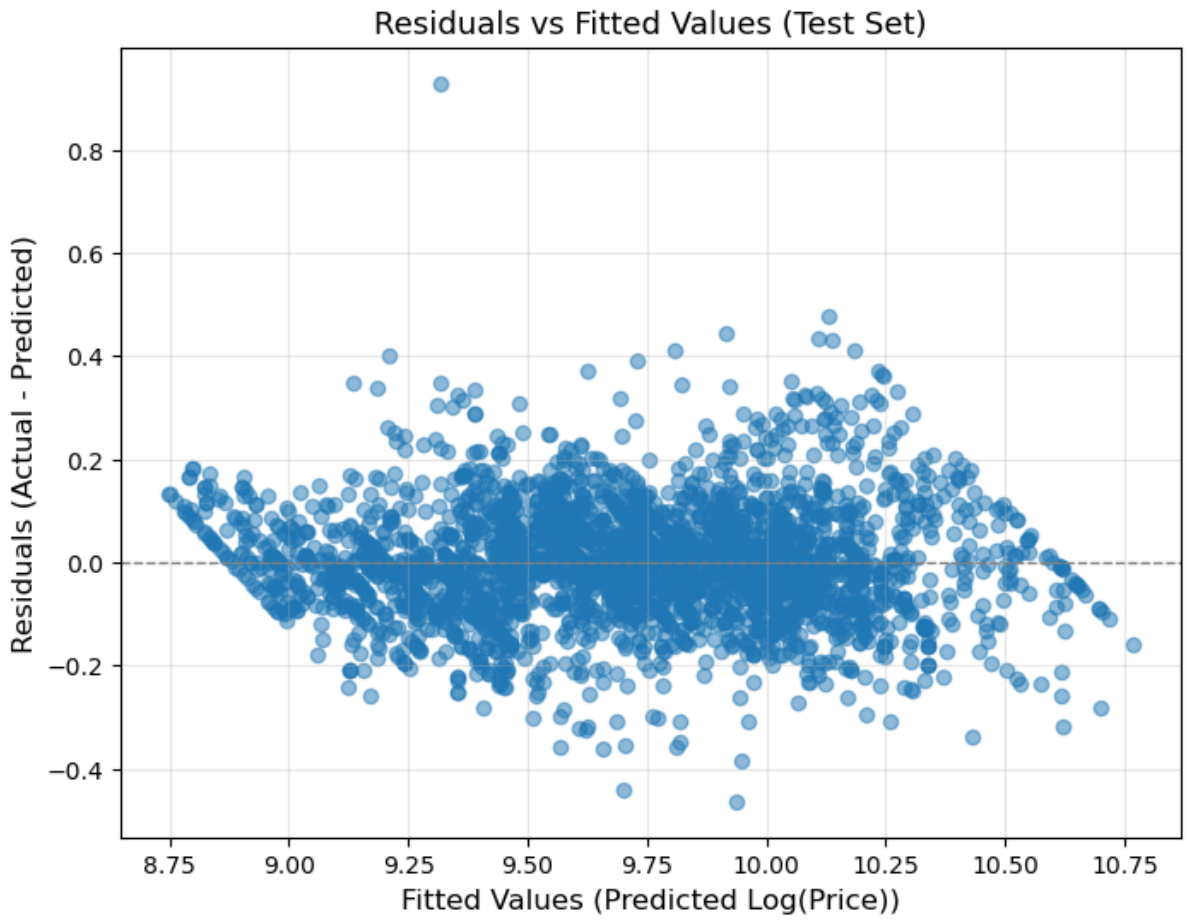


4. Model Building & Hyperparameter Tuning



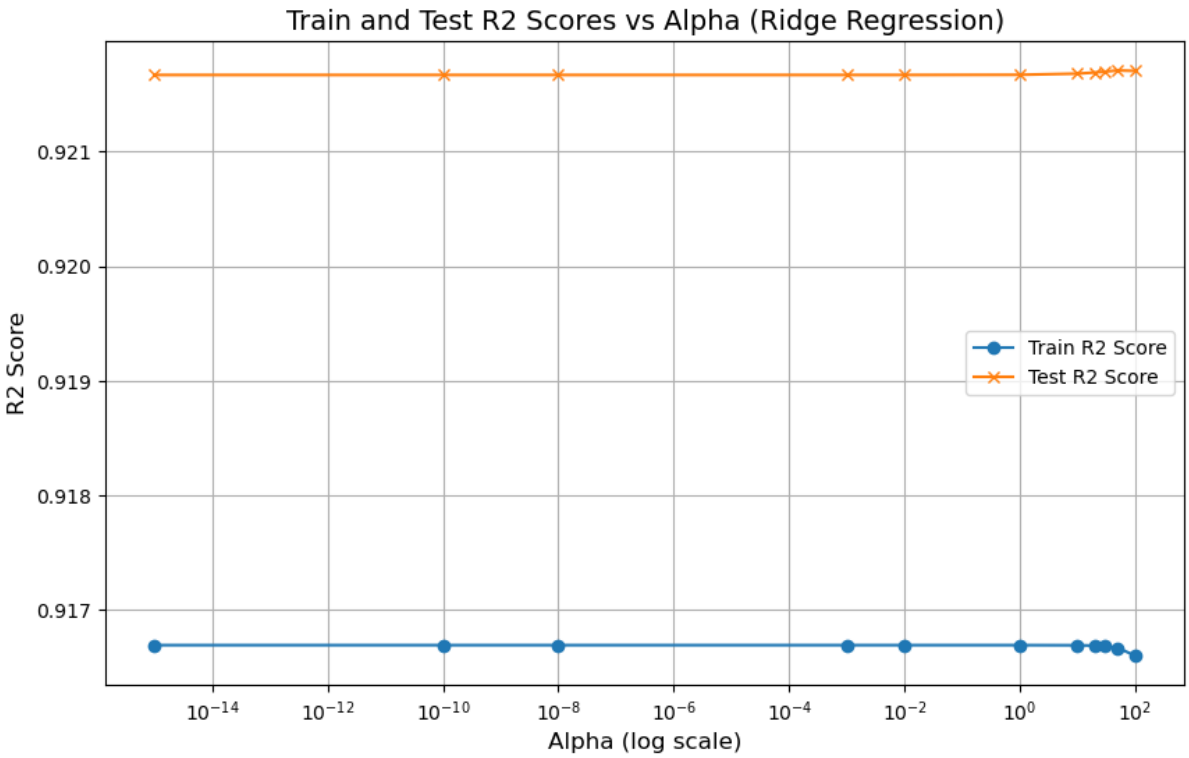


5. Model Performance Evaluation

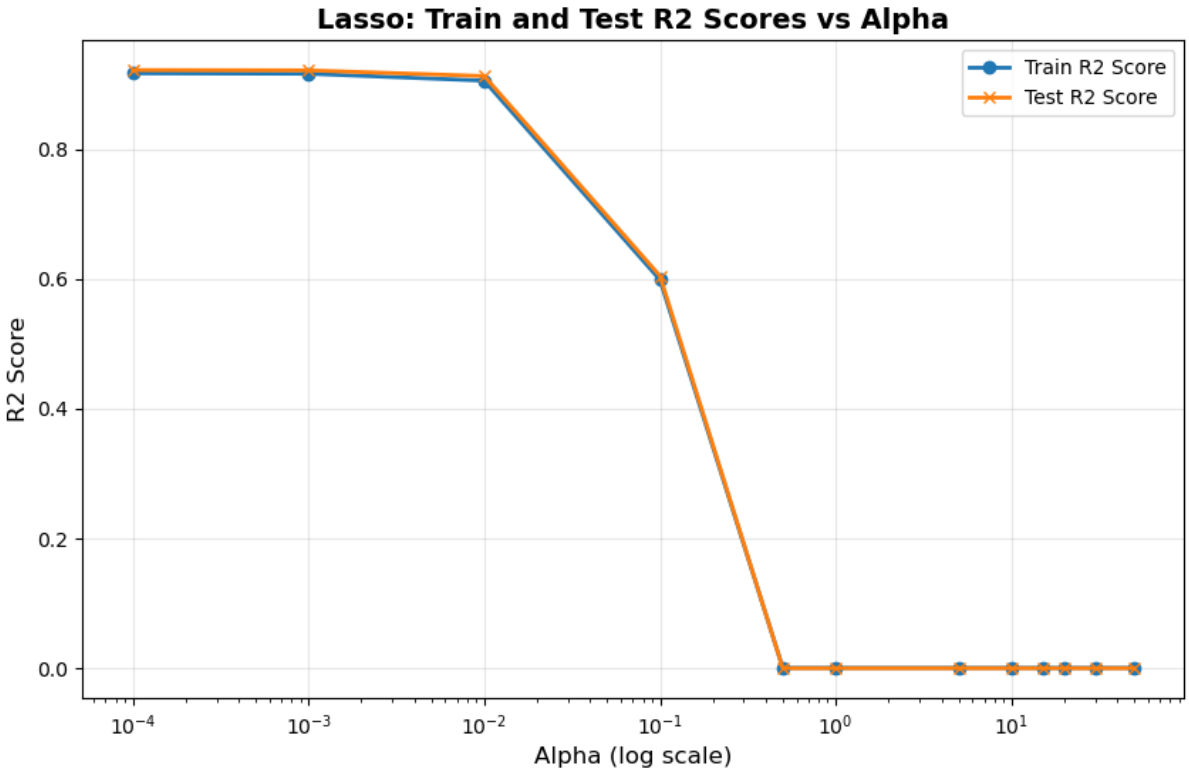


Interpretation:
- Points randomly scattered around 0 indicate good linearity.
- Patterns or curvature suggest model misfit or non-linearity.

✓ Linearity check visualisation completed



✓ Train and test scores plotted against alpha

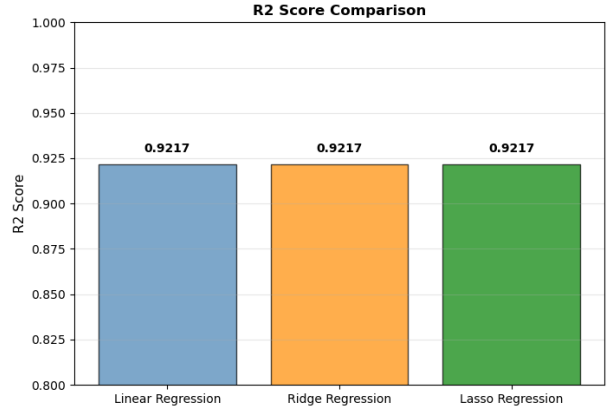


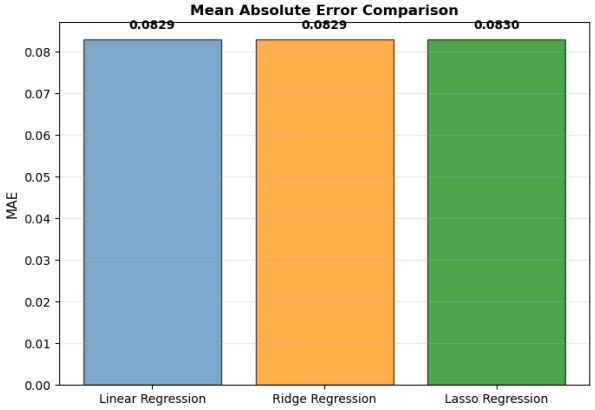
✓ Train and test scores plotted for Lasso regression

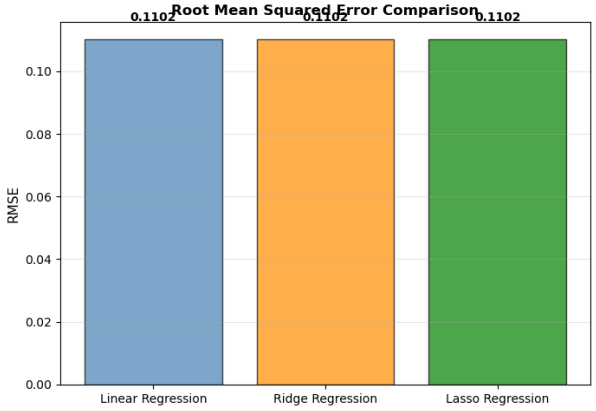
Detailed scores for each alpha:

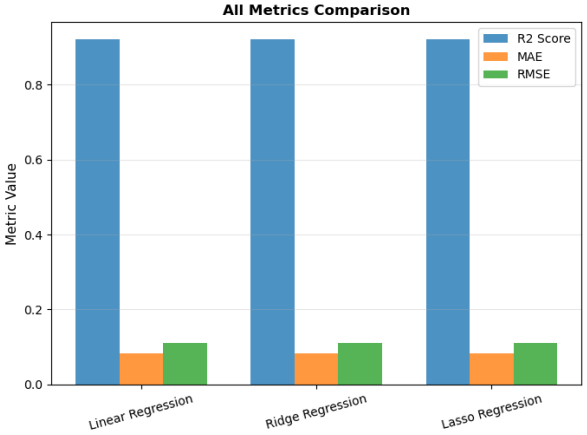
Alpha Train R2 Test R2 Difference

0.000100	0.9167	0.9217	-0.0050
0.001000	0.9161	0.9212	-0.0052
0.010000	0.9053	0.9124	-0.0071
0.100000	0.5982	0.6042	-0.0060
0.500000	0.0000	-0.0005	0.0005
1.000000	0.0000	-0.0005	0.0005
5.000000	0.0000	-0.0005	0.0005
10.000000	0.0000	-0.0005	0.0005
15.000000	0.0000	-0.0005	0.0005
20.000000	0.0000	-0.0005	0.0005
30.000000	0.0000	-0.0005	0.0005
50.000000	0.0000	-0.0005	0.0005

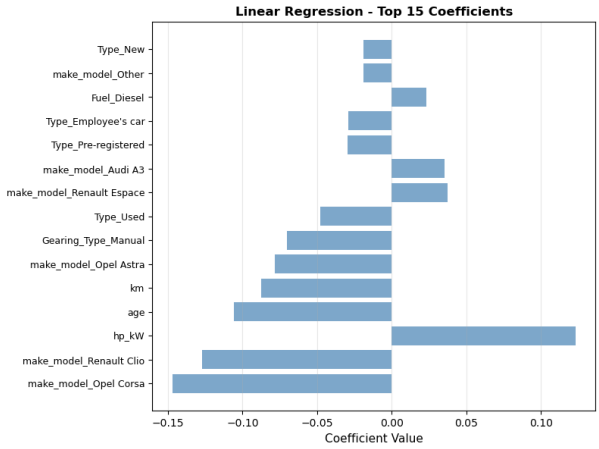


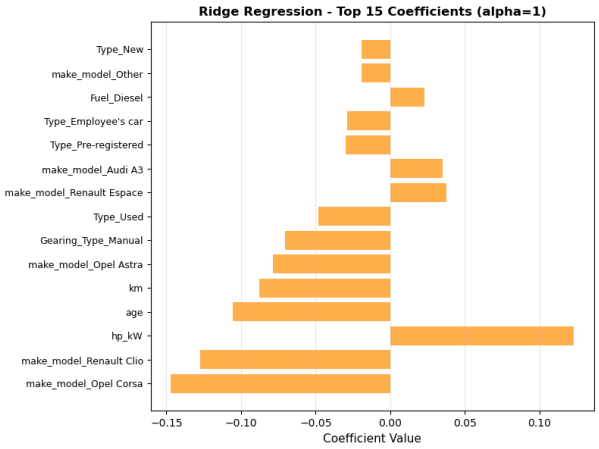


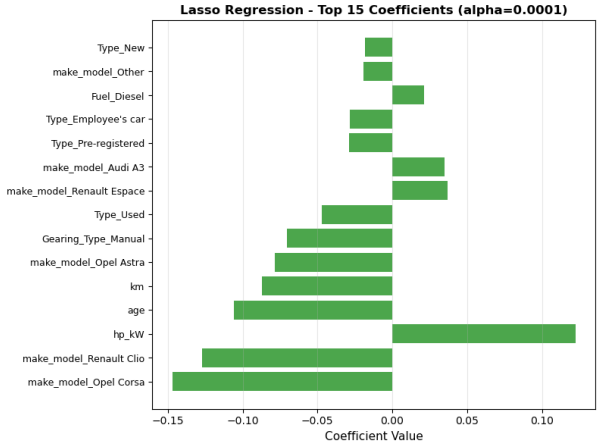


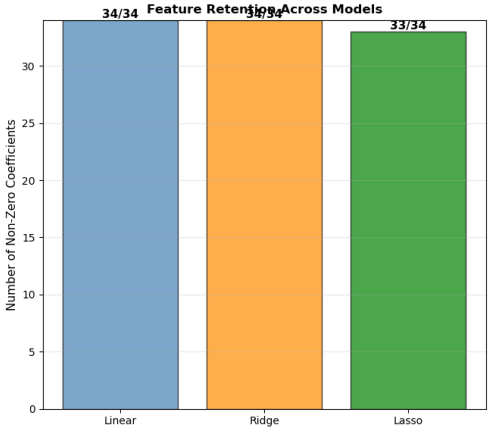


✓ Final model comparison completed









✓ Coefficient analysis completed